higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 3

(6050383)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 The Ritter liquor preservation process is used as a dry bulk bailing method to store bagasse.

1.2 A rotary vacuum washer is used to remove ink from secondary fibre.

1.3 The Kraft pulping process produces a stronger pulp than mechanically produced pulp.

1.4 The reason for removing cooking liquor out of cooked pulp is to recover expensive chemicals that were used during the pulping process.

1.5 Size and weight are the two most important quality requirements of finished paper.

(5 × 1) **[5]**

QUESTION 2

Give ONE word/term for each of the following descriptions. Write only the word/term next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 Anything that cannot be used to make pulp or is bad for the pulping process like stones and dirt.

2.2 A type of equipment that is used to break pulp down into single fibres.

2.3 The quantity or mass of a substance per unit volume, area or length.

2.4 A process that uses sodium chlorate, sulphuric acid and sulphur dioxide as raw materials for the production of chlorine dioxide.

2.5 The place on the calendar where the rolls or jaws are closer together.

(5 × 1) **[5]**

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QUESTION 3

Choose a/an item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.5) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	A type of chip screening equipment which operates in a circular motion	A	surfactants
3.2	A process in which particles are separated because of their ability to float on a particular liquid	B	web
3.3	Substance that helps to remove contraries like ink by making them separate easier in water	C	density
3.4	Bleaching equipment in which chlorine dioxide is stored	D	flotation
3.5	A continuous sheet of paper on a paper machine	E	dispersants
		F	flat gyratory screen
		G	calendar
		H	downflow tower
		I	upflow-downflow tower
		J	flat vibratory

(5 × 1) **[5]**

QUESTION 4

4.1 Logs that come into the woodyard are inspected while they are still on the trucks.

Name THREE things to be checked for during the inspection of the logs.

(Any 3 × 1) (3)

4.2 Logs should be kept wet during storage.

4.2.1 Name TWO reasons for keeping logs wet.

4.2.2 Give TWO methods that you can use to keep logs wet.

(2 × 2) (4)

4.3 Describe the first-in, first-out method of fetching logs from storage stacks. (1)

4.4 Describe THREE indicators of poor performance of a debarker. (3)

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4.5 How would you remedy the following poor performance of a wood chipper?

4.5.1 Knife on the disc have become dull

4.5.2 The chips are broken into pieces that are too small to be used for pulp

(2 × 1) (2)

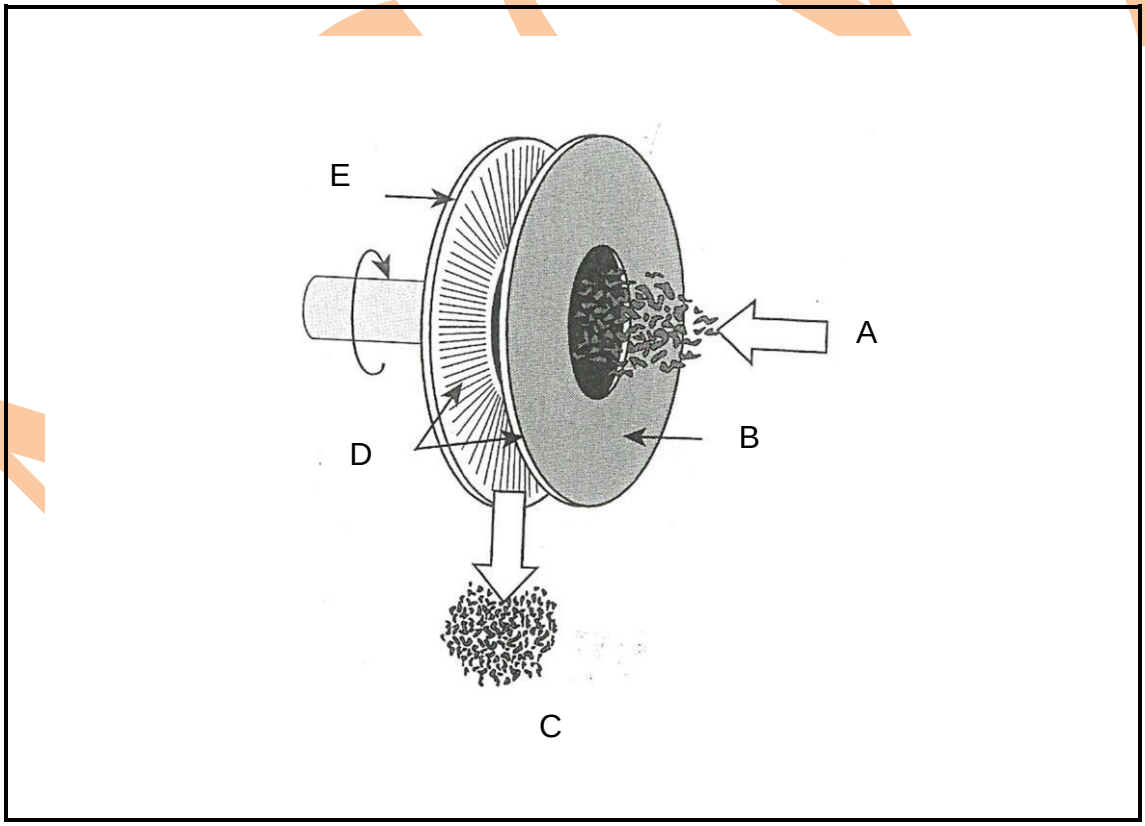
4.6 Name TWO actions you would take on the dewatering press, if the pith and sludge exiting the dewatering press are still very wet. (2)

4.7 Why is it important to remove contraries from the bagasse? (2)

[17]

QUESTION 5

5.1 Label the diagram below of a deflaking process by naming the parts (A–E) in the ANSWER BOOK.



(5 × 1) (5)

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5.2 What type of equipment will you use to conduct the following chemical tests on secondary fibre:

5.2.1 Cellulose solution viscosity

5.2.2 Conductivity

5.2.3 Shive count

(3 × 1) (3)

5.3 Explain the basic working principles of the flotation process of de-inking of secondary fibre. (7)

5.4 Pulpstone or grindstone is the main component in the grinding stone made from hard gravel stones of different sizes.

How would you remedy poor pulpstone operation indicated by the following:

5.4.1 Burn marks

5.4.2 Broken patterns

(2 × 1) (2)

[17]

QUESTION 6

6.1 Give reasons for the poor performance of a pre-steamer indicated by the following:

6.1.1 Low yield with high shive content

6.1.2 Fibres that are weak and brittle

6.1.3 Low tear and strength properties of fibres

(3 × 1) (3)

6.2 Name THREE advantages of the NSSC pulping process. (3)

6.3 Name THREE types of digesters used in the soda-pulping process. (3)

6.4 Draw a table indicating TWO differences between the Kraft pulping process and acid pulping process. (4)

6.5 Name FOUR examples of tissue paper produced from bagasse. (4)

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QUESTION 7

7.1 Pulp is treated with chemicals that make it whiter by bleaching.

7.1.1 Name THREE reasons for bleaching pulp. (3)

7.1.2 Name FOUR examples of bleaching chemicals used in the paper industry. (4)

7.2 Chlorine dioxide (ClO_2) is commonly used in the bleaching process.

Describe the production of chlorine dioxide using the hydrochloric acid base process. (4)

7.3 Name FIVE types of washers. (5)

7.4 What causes the spent liquor to be too diluted? (1)

[17]

QUESTION 8

8.1 Embossing is a finishing process that produces patterns or images on the surface of the paper.

8.1.1 Give TWO reasons for embossing. (2)

8.1.2 Draw a labelled diagram of the embossing process. (5)

8.2 What are the main functions (uses) of the following equipment used during the lamination process:

8.2.1 Spreader rolls

8.2.2 Press rolls

(2 × 1) (2)

8.3 What is the main cause of web breakage in the coater? (1)

8.4 What are *agitators*? (2)

8.5 How would you remedy poor finishing performance if there is too much or too little paper in a ream? (1)

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8.6 Define the following piling systems:

8.6.1 Laminar flow effect

8.6.2 Jagging

(2 × 2) (4)
[17]

TOTAL: 100

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